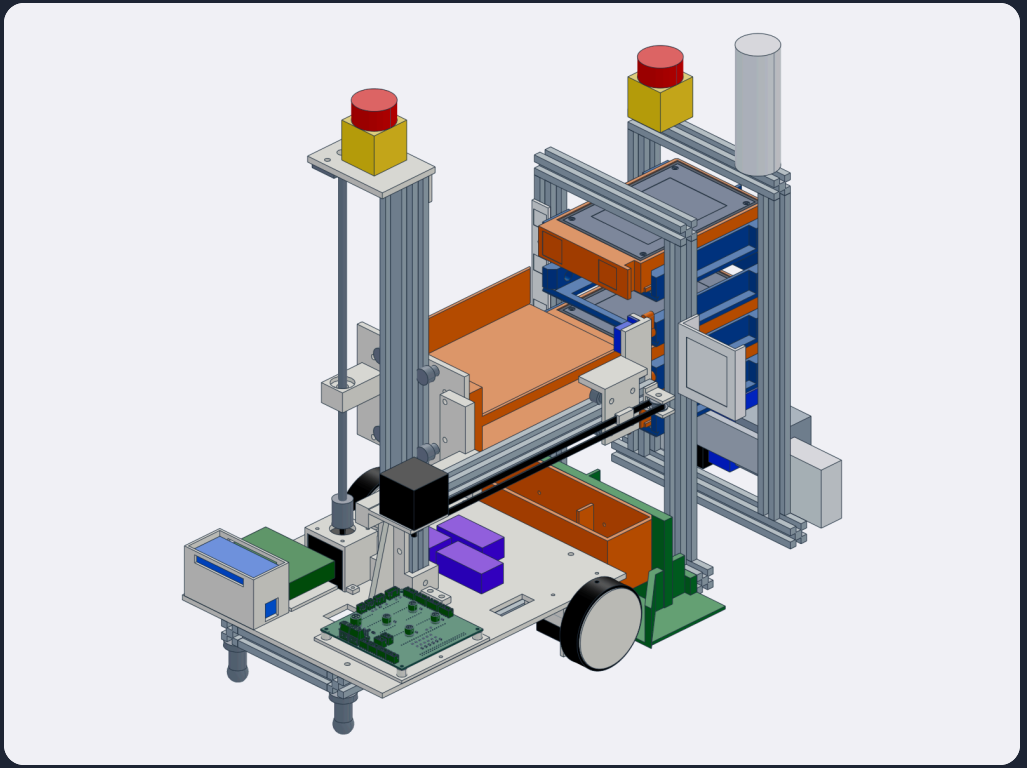
DELTA CUP 2026 · CORVIN / RACKBOTGEN3

Prototype review

Judged as a **working competition prototype**, not an industrial product.
Mechanical · 3D print · Electrical · Software · Parts · Timeline · Pitch

Deadline: 12 Nov 2026 (fly) · Final 14 Nov



Review date 23 Oct 2026 · repo `main @ d776010`, spec v0.1.27

Bottom line

20 days
to fly. Nothing built yet.

1 / 40
parts ordered (PCB only). Rest 15 days late.

≈ 548 $
budget vs 500 limit. Realistic ≈ 595.

≈ 165 h
of 3D printing still to do.

Verdict (prototype bar): the concept makes sense. Materials are right, the power budget is fine, the PCB is clean and ordered, and the CAD passes 41/41 clearance checks. What can still sink the demo is **a handful of geometry bugs plus the schedule**, not the architecture.

Would break the demo: fix these

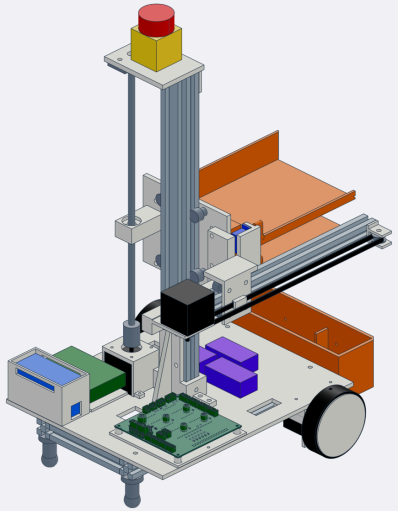
- **Arena 700 mm is too short** for the turn; make it ≥ 900 mm.
- **Approach code ignores the camera offset**, so the robot docks off-centre.
- **Dock yaw** up to 7° vs 0.7° the extraction tolerates.
- **Y-carriage arm joint is too thin**, and the **belt clamp has no screw** (CAD).
- **Z offset from the bay tag is a placeholder**, so the lift can't line up.

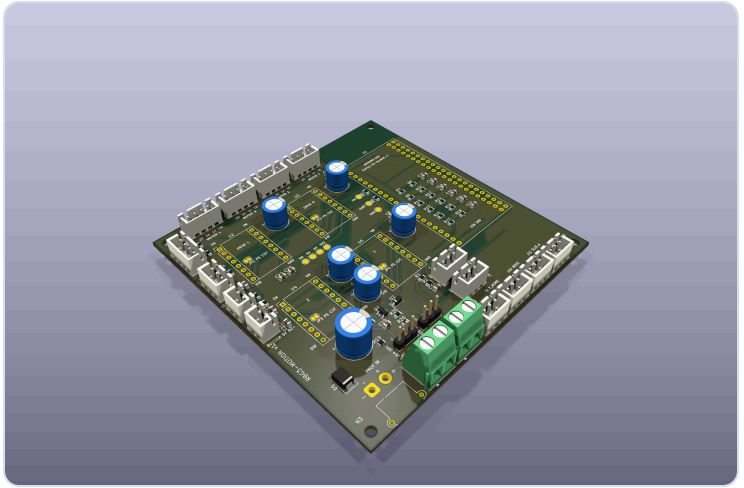
Do this week

- Order **every** part 23–24 Oct, plus **+1 kg PETG**.
- Get a **2nd printer** / print service (≈ 165 h of prints).
- Choose the **ESP32 rack controller**; the PLC is a bonus.
- Fix the 5 items on the left **before printing** the robot parts.
- Fix the pitch: budget, PLC and router claims.

Status at a glance

AREA		DONE	MISSING
Mechanical / CAD	🟠	All parts modelled. robot_check 41/41, station 14/14	Fit test T1 never run. No yaw/tolerance check. bay_fit check crashes (STEP not built)
Motion PCB	🟢	Ordered 23 Oct (JLC, 2 assembled). Review fixes in: C18–C23, R25, TP1–7	Arrives ≈ 30 Oct–3 Nov. THT solder + bring-up
Robot power	🟠	Spec + SPICE done	Battery, BMS, relay, bucks not bought. Inrush not bench-tested
Pi software	🟠	Mission, Modbus client, SQLite, dashboard (sim)	Never run on hardware. Vision bugs. No calibration tools
ESP32 firmware	🟠	Core + HAL + 20 unit tests	Not on hardware. No host watchdog
Rack controller	🔴	Spec only	No firmware, no ladder. PLC loan undecided
Parts	🔴	List complete-ish	39 of 40 rows not ordered, ≈ 10 items missing
Pitch	🟠	13-slide deck + 90 s video	Stale (v0.1.22). Budget and PLC claims wrong





What doesn't make sense (even for a prototype)

#	HOLE	WHY IT BREAKS THE DEMO	CHEAP FIX
1	**Arena too short**	Stations are 700 mm apart and the robot is 360 mm long with a 283 mm turn radius. After undocking it clears the other dock by ≈ 2 mm and has ≈ 40 mm left to correct. The sim uses 1500 mm, so the 107 s cycle time is not valid. *Two reviewers found this separately.*	Stations ≥ 900 mm apart, undock 200 mm, tape marks on the mat
2	**Approach aims at the wrong line**	The code steers the tag's camera-frame offset to 0. The tag is on the right upright and the camera is 78 mm left, so the robot ends ≈ 150–200 mm off the dock axis. The sim hides it (tag dead ahead).	One offset per station in `config.yaml`
3	**Dock yaw vs extraction**	The funnel turns sideways error into yaw (up to ≈ 7°). The caddy channel tolerates ≈ 0.7°. DOCK stops the instant NOSE closes, so the nose never squares up.	Push ≈ 3 mm more at low current after NOSE
4	**Y carriage: weak arm + belt clamp only half modelled**	The arm joins the frame on only 33 mm²: 10 N at the finger gives ≈ 20 MPa at a sharp corner, so a jam snaps it. The clamp comment promises a screw + GT2 teeth, but the model is a plain slot, so the belt slips.	Add a web under the frame (→ 290 mm²) + an M3 through the jaw. Cap the Y current
5	**Z offset from the bay tag is a placeholder**	`shelf_z_error` returns the raw pose with no reference, so the lift never lines up with a bay (E_ALIGN every time).	Measure once per bay; store fixed Z offsets
6	**Rack controller doesn't exist**	The pitch shows a PLC; the loan is unlikely. No ESP32 firmware or ladder logic is written. Without it: no lamps, no dock relay, no bay switches.	Write the ESP32 Modbus server (1–2 days)

Prototype lens: what you can safely ignore

OK FOR A PROTOTYPE skip or note in the pitch

- Open Wi-Fi / no dashboard login: WPA2 password is enough.
- Pi single point of failure: restart it by hand if it crashes.
- SMART / "IIoT" data simulated: just say "simulated".
- No reverse-polarity FET: keyed XT30 + fuse is fine.
- WS2812 at 3.3 V: a status light only.
- 50-cycle reliability: 20 clean cycles is a strong demo.
- Energy logging (INA226), auto-charging on the dock.
- Digital twin, MQTT, HMI, caddy QR, cloud hosting.
- Nominal-only CAD checks: tune on the real parts.

DON'T IGNORE cheap, and they bite on stage

- Robot E-stop relay works (the only hard stop): test it.
- ESP32 stops motors if the Pi goes silent (≈ 10 lines of code).
- USB serial reconnect: a loose cable mid-demo = dead robot.
- Don't re-home Z blind after a reset: check slots first.
- Inrush on E-stop release: try it ×20 on the bench (BMS trip).
- Keep the robot's GND off the dock frame (charger limit).
- Pitch numbers must match reality (budget, PLC).

Mechanical: function and fit

	ISSUE	EVIDENCE	PROTOTYPE FIX
FIX	Front caster height wrong in docs	Docs say 27 mm, the CAD needs 23.5 mm. At 27 mm the ball rides the 3 mm dock plate and lifts the drive wheels. `parts.csv:9`, spec 01:243	Spacer for 23.5 mm
FLAG	Finger engagement ±1 mm	Worst-case stack-up ≈ ±2.5 mm; notch chamfer is only 0.5 mm	2 mm chamfers; tune `y_pull` per station
FLAG	Push shoves the robot back	Drives hold at 50 % (≈ 6 N); a push needs 3–5 N + stiction	Hold at run current during Y
FLAG	Z torque at 15 mm/s	450 rpm on 12–16.8 V is marginal; skipped steps aren't detected	Bench test with 2 kg at 13 V; slow down if needed
FLAG	4 rigid contact points	A 0.5 mm spacer error unloads a drive wheel	Shim the casters at assembly
FLAG	Cable routing not designed	Servos, switches, the Y motor and CSI ribbon ride 193 mm of Z + 180 mm of Y	Spiral wrap + zip ties, leave slack loops
IMP	Funnel walls low; parts located by T-nut	The 40 mm wall overlaps the nose by only 10 mm	Walls ≈ 70 mm; mark positions with tape

CAD checks: robot 41/41, station 14/14 pass. Mass 4.39 kg, tip-over margin 69 mm vs 6 mm needed. Nominal geometry only, so tune on the real parts.

3D print: materials & setup

Materials are correct. Every robot part is PETG, and no PLA sits near a motor (NEMA17 runs 50–70 °C, PLA creeps above ≈ 55 °C). PLA is used only for dummy drives and tags. The TPU tire is on a PETG hub. Every STEP file is one valid solid, in mm.

GROUP	MATERIAL	SETTINGS
Loaded robot parts (brackets, collar, carriages, motor mounts, hub)	PETG	0.2 mm, **5 walls**, 40 % gyroid, 240–250 °C
Fingers	PETG	100 % infill, flat
Big parts (tray, nose, shelves, caddies, dock)	PETG	4–5 walls, 15–20 % gyroid
Dummy drives, tags	PLA	3 walls, 15 %
Tires	TPU 95A	20 mm/s, 100 %, direct drive

Problems in the CAD / STL setup

- **STLs come out in assembly pose**, not print pose. Rotate each in the slicer: plates face down, tray on its back, fingers/idler on the y-face, hub outboard face down.
- **8 STLs have small open edges** (deck, carriages, caddy, shelf, tags, dummy). Run slicer repair. Check the dummy's 3 insert bosses are still there.
- **Deck 246 × 300 mm** won't fit a 220 bed and may warp. Cut it from plywood/acrylic using `deck.dxf`.
- **Dock block 298 mm** also needs a big bed, or split it.
- **Caster spacers not modelled**: measure the casters, then model and print at 100 %.
- Fit coupon doesn't test the 7.3 mm eccentric, 22.5/23 mm boss or 2.2 mm pilot holes. Add them.

Filament: PETG ≈ **2.6–2.9 kg needed, only 2 kg ordered** → buy +1 kg (or plywood deck −0.23 kg). PLA 0.7 of 1 kg OK. TPU 0.05 kg. **≈ 165 h printing** (robot ≈ 50 h).

3D print: structural check (hand calcs, safety factor ≥ 2)

PART	VERDICT	WHY (NUMBERS)	CHEAP FIX
y_carriage arm	WRONG	Joint only 33 mm². 10 N at the finger → ≈ 20 MPa at a sharp corner (PETG layers ≈ 25 MPa). A Y stall (30–40 N) snaps it.	Web x 76.5–82 under the frame → 290 mm²; cap Y current ≈ 300 mA
y_carriage clamp	WRONG	Comment says screw + GT2 teeth; the model is a plain 1.9 mm slot for a 1.38 mm belt, so it slips.	M3 through the jaw (zip-tie for now)
finger ×2	WEAK	Only a Ø2.2 hole, no horn pocket, so it can spin on the servo. Bending is fine (1.5 MPa).	Horn pocket + 2 screw holes
tray_stack	WEAK	Stress < 1 MPa, but 4 mm floors with a free edge sag ≈ 1 mm under 1 kg, which equals the ±1 mm Z tolerance.	10 mm lips or 5–6 mm floors; tune Z offsets
rail_bracket	WEAK	4 mm motor plate sticks out 52 mm; belt twists it ≈ 0.8°. Boss hole 22.5 prints tight.	6 mm plate + rib; hole 23
y_idler_bracket	WEAK	3 mm fork arms carry 20–40 N of belt tension constantly: 6–12 MPa → creep.	5 mm arms
mast_collar	OK*	Gusset ≈ 3 MPa. Holds only if the 2 M5s stay tight; if loose, the mast rocks ≈ 1° (tray front drops 3–4 mm).	Washers, thread-locker, re-torque after day 1
drive_bracket ×2	OK	2.5–11 MPa, deflection ≈ 0.15 mm	Print plate face down
z_motor_mount, top plate, z_carriage	OK	≈ 1–4 MPa normally. A crash at stall ≈ 36 MPa in the mount feet.	Keep the home switch + current limit
drive_hub / tire	OK	Set screw in a brass insert; D-flat ≈ 7 MPa	Thread-locker
battery box, nose, caddy, shelf, dock	OK	Low loads (the 4 × 18650 pack is ≈ 0.25 kg)	—

Electrical / PCB

DONE **Schematic review applied, verified in the netlist**

- +6V servo: C18 220 µF/25 V + C19 100 nF
- +5V: C20 10 µF + C21. +3V3: C22 10 µF + C23
- R25 10 k pull-down on the buzzer base
- TP1–TP7 test pads (VM, 5V, 6V, 3V3, UART, EN, GND)
- Robot E-stop NO contact → J13 (instant detect)
- Pin map, UART addresses, dividers, flyback: all OK

ORDERED **JLCPCB 23 Oct**: 5 boards, 2 with SMD assembly, FedEx, 32 $

FLAG **Still open**

- **Import duty** ≈ +10–13 $ on the PCB (not budgeted). ETA 30 Oct–3 Nov.
- **Inrush** 120–220 A on every E-stop release can trip the BMS. Bench test ×20.
- **Charger grounding**: if robot GND ever touches the dock frame, the XL4015 current limit is bypassed.
- **WS2812 at 3.3 V**: out of spec. Buy an SK6812 if it flickers.
- **J2/J3 screw terminals** aren't keyed. Check polarity with a meter at bring-up.
- **Fallback** if the PCB is late: perfboard parts are not on the list. The 3 bare boards need SMD hand-soldering.

Power budget OK: peak ≈ 30 W (2.2 A) vs 7.5 A fuse → 3× margin. 50 Wh ≈ 2.5 h.

Software / IoT

Flow: dashboard (FastAPI on the Pi, which is also the Wi-Fi AP) → mission state machine → Modbus TCP over Wi-Fi to the rack controller (lamps, dock relay, 2 s watchdog), and USB serial to the ESP32-S3 (motors, servos, interlocks). Localization: open-loop undock + 180° turn → AprilTag approach → blind push into the funnel → NOSE switch.

	BUG / GAP
FIX	Approach ignores the tag/camera offset (slide 4 #2)
FIX	`shelf_z_error` is a placeholder (raw pose, no reference) → E_ALIGN on every bay
FIX	Rack E-stop is never read
FIX	USB serial: no reconnect, BOOT event ignored, CFG never sent
FIX	Startup re-homes Z before checking for a caddy straddling tray and bay
FIX	`wait_for` ignores `online`; rack heartbeat D2 is never checked
FLAG	Docking margin ≈ 17 mm vs ±20 mm funnel; ψ is noisy; no tag search
FLAG	ESP32 Wi-Fi power-save can trip the 2 s watchdog (1 line to disable)
IGNORE	Dashboard security, Pi redundancy, simulated SMART data: fine for a prototype

COMPONENT	STATE
Pi app (mission, Modbus, SQLite, dashboard)	sim only
Motion firmware + 20 unit tests	not on HW
AprilTag vision	untested + bugs
Calibration / tag tools	missing
Camera stream	stub (503)
Rack controller (ESP32 fw or ladder)	missing
Pi AP / systemd / deploy	missing
Hardware integration	not started

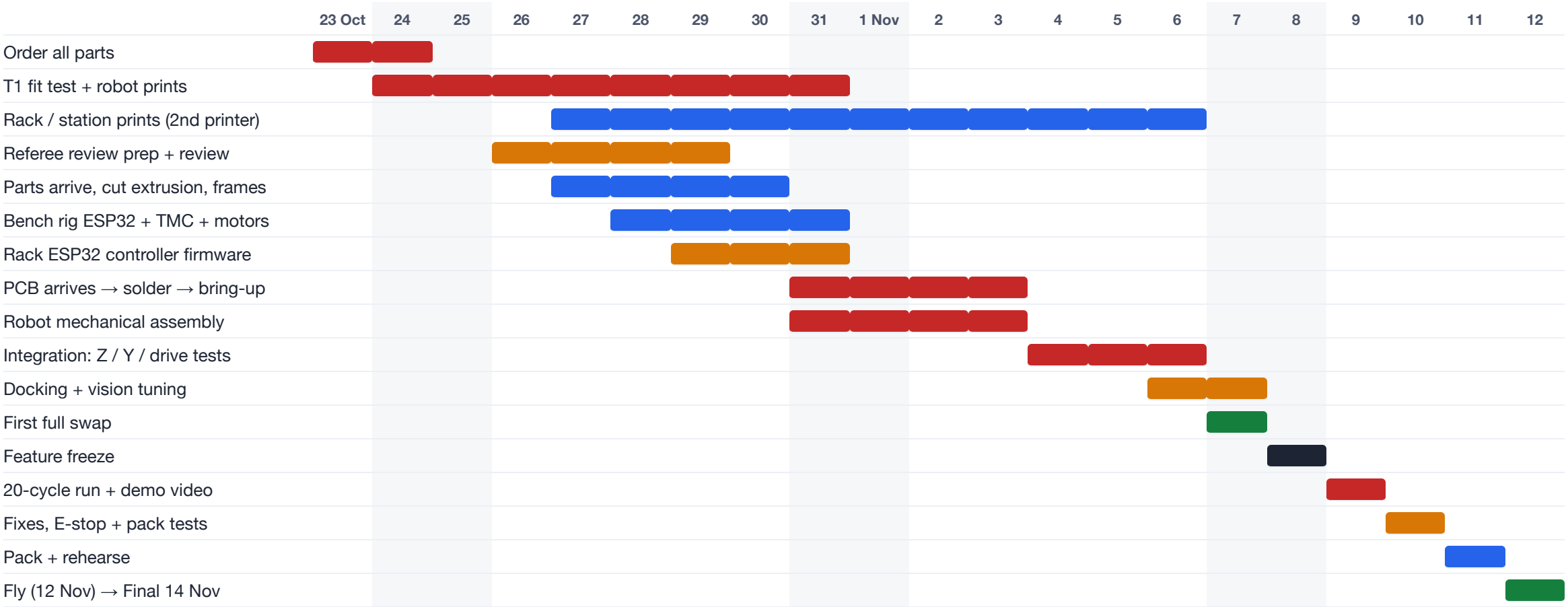
Parts & budget

ID	PROBLEM	ACTION
All but P22	Not ordered; `order_by` was 8 Oct	**Order 23–24 Oct**
P25 PLC	Loan decision overdue since 13 Oct	Commit to the ESP32 (P40)
P21 E-stop	Needs 1NO + 1NC; the backup listing is 1NC only	Buy LA38-11ZS 1NO+1NC
P16 ESP32	Only one, no spare	+1 (+7 $)
P17 TMC2209	csv says 4, the shopping list buys 5	Set qty 5
P11	No M5×30 V-wheel axles or nylocks (×8); M3 for the belt clamp	Add
P32 PETG	2 kg ordered, ≈ 2.6–2.9 kg needed	+1 kg (≈ +12 $)
Missing	Pi micro-USB pigtail, drag chain, motor/servo extensions, JST crimper, heat-shrink, Pi heatsink, tether lead	Add ≈ +25 $
P26	Polyfuse question open	UK5-HESI + 3.15 A fuse
Docs	BACKLOG / overview say ≈ 480 $; P10 count mismatch	Update

BUDGET	USD
Mechanical	123
Robot electronics	248
Rack electrical	77
Printing & demo	100
Spec total	**548**
+ duty, missing, spares, PETG	≈ +50
Realistic	**≈ 595–600**
Limit	500

Savings: team-owned filament / fasteners –20 to –50 · drop 3rd servo –4 · OV5647 camera –29 (risky) · drop Claude Pro line –23 (referee may question it).

Timeline 23 Oct → 12 Nov (critical path, slack ≈ 0–1 day)



red critical path · **amber** at risk · **blue** parallel · **green** milestone. First full swap moves from the plan's 1 Nov → ≈ 7 Nov. Feature freeze 2 Nov → 8 Nov.

MVP: cut to make 12 Nov

KEEP must work on stage

- One full swap: dock → pull failed caddy → fetch spare → push in
- AprilTag approach + funnel docking + NOSE confirm
- Robot E-stop (hard cut) + dock interlock
- Rack controller (ESP32) with lamps + bay switches
- Dashboard: inject fault → job → live state
- Delta PSU on the rack ("Delta inside")

CUT / STRETCH

- Two-slot tray → **single slot**, two trips
- Populate **2 bays + 2 spare slots** (4 caddies, 4 dummies)
- Dock auto-charge → **charge manually**
- INA226 energy logging + shunt swap
- Tag fine-alignment (T7) → fixed Z offsets + funnel
- 50-cycle test → **20 cycles**
- Digital twin, MQTT, HMI, caddy QR/ID
- PLC loan: bonus if it arrives, not on the critical path

> **Rule:** record the demo video the day the first swap works (≈ 7–9 Nov), not at the end. That is your insurance.

Pitch deck: fix before 27/29 Oct

SLIDE	WRONG / OVERCLAIM
1	"[27 / 29]" placeholder; footer v0.1.22 (now v0.1.27)
3, 13	Travel router + PLC shown as certain; router dropped, PLC unlikely
4 / video	Slide says 107 s, video clock 2:45; video shows the old workflow
8	Cloud hosting not in spec; "≥ 8 fps" untested; table names differ
9	"Built with…" but nothing is built; 39 SMD → 45
12 / video	**"≈ 480 $ / < 500 $"**; real ≈ 548–600
12	"≤ 20 Oct done" overstated

LIKELY JUDGE QUESTION	ANSWER
Why a robot? A tech swaps in 2 min.	Lights-out / edge sites: truck roll = hours, RAID degraded. MTTR + cost per swap.
Scale to 19" 42U racks?	Pitch and Z are parameters; aisle rail + latch actuator in production.
Where is Delta, really?	State the PLC status honestly. Roadmap: Delta servos + HMI.
Nothing built: why believe it?	42-pose CAD check, SPICE, sim, dated hardware gates.
Over budget?	Show the savings plan.

Action list

#	ACTION	AREA	BY
1	Order all parts + missing items (pigtail, drag chain, extensions, crimper, M5 axles, spare ESP32/TMC)	Parts	**24 Oct**
2	Line up a 2nd printer / print service; start T1 fit test, then robot parts first	Mech	**24 Oct**
3	Decide ESP32 rack controller; update 00/03 spec + deck	System	**24 Oct**
4	Fix pitch deck (budget, PLC, router, 107 s vs 2:45, placeholders)	Pitch	**26 Oct**
5	Arena ≥ 900 mm, undock 200 mm; camera→dock offset in approach + sim	SW	27 Oct
6	Rack-E-stop handling, ESP32 host watchdog, serial reconnect, safe startup	SW / FW	30 Oct
7	Write the rack controller firmware (Modbus server, watchdog, lamps, relay)	FW	31 Oct
8	CAD before printing: y_carriage web + clamp screw, finger horn pocket, tray lips, rail/idler bracket thickness, caster 23.5 mm	Mech	**before printing**
9	Bench rig: ESP32 + TMC + motors; Z torque at 13 V; Y pull force	HW	31 Oct
10	PCB bring-up per README; inrush ×20 on E-stop release	HW	3 Nov
11	Docking yaw fix (seat push + yaw check); `shelf_z_error` calibration	SW + Mech	6 Nov
12	First swap → record video same day; 20-cycle run; pack test	All	9 Nov